HastaDAQ

Net labels connect everything: same name = same net. +3V3 / GND are plain labels, not power symbols.

Pimoroni Pico Plus 2 (+ RM2 on SP/CE): same 40-pin header as Pico 2 W. I2C0 = GP4/GP5 (header pins 6/7), to the IMU Qwiic-breakout header and to the 4-wire loom for the strain-gauge board. Crossed-out pins are intentionally unused.

J1
IMU via Qwiic breakout (The Pi Hut 106676)

GND 1 — GND
+3V3 2 — 3V3
SDA 3 — SDA
SCL 4 — SCL

J2
GAUGE BOARD LOOM (J_I2C): GND, +5V, SDA, SCL

GND 1 — GND
+5V 2 — 5V
SDA 3 — SDA
SCL 4 — SCL

Strain-gauge board: 4-wire I2C loom. Pin 2 is +5V, NOT 3V3 -- use a keyed plug, never a Qwiic cable. Pull-ups live on the ADS122C04 breakouts on that board.

U1
Pimoroni Pico Plus 2 (header pins)

SBUS_IN_GPIO	2	GP0_UART0-TX	VBUS	40	
GND	3	GP1_UART0-RX	VSYS	39	+5V_VSYS
	4	GND	GND	38	GND
	5	GP2	3V3_EN	37	
SDA	6	GP3	3V3(OUT)	36	+3V3
SCL	7	GP4_I2C0-SDA	ADC_VREF	35	
GND	8	GP5_I2C0-SCL	GP28	34	
	9	GND	AGND	33	GND
ENC4_CS	10	GP6	GP27	32	
ENC3_CS	11	GP7	GP26	31	
ENC2_CS	12	GP8	RUN	30	
GND	13	GP9	GP22	29	
SPI_SCK	14	GND	GND	28	GND
SPI_MOSI	15	GP10_SPI1-SCK	GP21_UART1-RX	27	SBUS
SPI_MISO	16	GP11_SPI1-TX	GP20_UART1-TX	26	SBUS
ENC1_CS	17	GP12_SPI1-RX	GP19	25	
GND	18	GP13_SPI1-CSn	GP18	24	
TRIG_IN_GPIO	19	GND	GND	23	GND
TRIG_OUT_GPIO	20	GP14	GP17	22	
		GP15	GP16	21	SOFTSTOP_IN

J3
ENC1 SPI port: MT6835 (pitch) / Orbis

GND 1 — GND
ENC1_CS 2 — CS
SPI_SCK 3 — SCK
SPI_MOSI 4 — MOSI
SPI_MISO 5 — MISO
+3V3 6 — 3V3
+5V 7 — 5V
GND 8 — GND

J4
ENC2 SPI port: MT6835 / Orbis

GND 1 — GND
ENC2_CS 2 — CS
SPI_SCK 3 — SCK
SPI_MOSI 4 — MOSI
SPI_MISO 5 — MISO
+3V3 6 — 3V3
+5V 7 — 5V
GND 8 — GND

J5
ENC3 SPI port: MT6835 / Orbis

GND 1 — GND
ENC3_CS 2 — CS
SPI_SCK 3 — SCK
SPI_MOSI 4 — MOSI
SPI_MISO 5 — MISO
+3V3 6 — 3V3
+5V 7 — 5V
GND 8 — GND

J6
ENC4 SPI port: generic SPI device

GND 1 — GND
ENC4_CS 2 — CS
SPI_SCK 3 — SCK
SPI_MOSI 4 — MOSI
SPI_MISO 5 — MISO
+3V3 6 — 3V3
+5V 7 — 5V
GND 8 — GND

CS pull-ups (10k)

+3V3 — R1 10k — ENC1_CS
+3V3 — R2 10k — ENC2_CS
+3V3 — R3 10k — ENC3_CS
+3V3 — R4 10k — ENC4_CS

U2
Pololu D42V55F8 8.4V 5A buck (servo rail)

VSERVO	1	VOUT	VOUT	2	VSERVO
GND	3	GND	GND	4	GND
GND	5	GND	GND	6	GND
+12V	7	VIN	VIN	8	+12V
	9	VRP	VRP	10	
	11	PG	EN	12	

U3
SparkFun BabyBuck 5V (AP63357) (Pico VSYS, Orbis)

+12V	3	VIN	VOUT_5V	1	+5V
GND	2	GND			
	4	EN			

D1
Schottky (e.g. 1N5822, 3A)
+5V — A — K — +5V_VSYS

C1
100uF 25V input bulk (Panasonic ECA1EM101)
+12V — VSERVO_GND

C2
470uF 16V servo bulk (Panasonic EEUFR1C471)
VSERVO_GND

C3
100uF 10V 5V bulk (RS PRO LHK 5x7)
+5V — GND

R5
1k (soft-stop series)
SOFTSTOP_CABLE — SOFTSTOP_IN

R6
10k (soft-stop pull-up)
+3V3 — SOFTSTOP_IN

D2
5.1V zener (BZX55C5V1) soft-stop clamp
GND — A — K — SOFTSTOP_IN

J7
SBUS servo header 3x6 (6 plugs: GND, V, SIG)

GND	1	S1_GND	
VSERVO	2	S1_V	
SBUS	3	S1_SIG	
GND	4	S2_GND	
VSERVO	5	S2_V	
SBUS	6	S2_SIG	
GND	7	S3_GND	
VSERVO	8	S3_V	
SBUS	9	S3_SIG	
GND	10	S4_GND	
VSERVO	11	S4_V	
SBUS	12	S4_SIG	
GND	13	S5_GND	
VSERVO	14	S5_V	
SBUS	15	S5_SIG	
GND	16	S6_GND	
VSERVO	17	S6_V	
SBUS	18	S6_SIG	

TEST POINTS (plain 2.0 mm pads): two GND pads for a scope clip (near the supply and near the Pico). Rails and signals are probed at component leads and header pins (C1/C2/C3, D1, Pico pins 36/39).

GND — TP1 TP
GND — TP2 TP

PWR_FLAG: marks nets supplied through the supply connector (+12V, GND) and the Schottky (+5V_VSYS)

#FLG1 PWR_FLAG — +12V
#FLG2 PWR_FLAG — GND
#FLG3 PWR_FLAG — +5V_VSYS

J10
SBUS IN (RC receiver): GND, V, SIG

GND 1 — GND
VSERVO 2 — V
SBUS_IN 3 — SIG

SBUS_IN — R13 1k series — SBUS_IN_GPIO
SBUS_IN_GPIO — R14 10k pull-down — GND

SBUS IN: RC-receiver pass-through (plug order GND, V, SIG). V = VSERVO 8.4 V -- check the receiver's max supply. SIG -> 1k -> GP1 (UART0 RX), 10k to GND.

J8
Main supply cable (2x4): VSUPPLY, VGND, TRIG, SOFTSTOP

+12V 1 — VSUPPLY
GND 2 — VGND
TRIG_IN 3 — TRIG_IN
SOFTSTOP_CABLE 4 — SOFTSTOP+
5 — SOFTSTOP−
TRIG_OUT 6 — TRIG_OUT
GND 7 — VGND
+12V 8 — VSUPPLY

TRIGGER IN: any 3.3 V .. 24 V level, opto input

TRIG_IN — R7 10k pull-down — GND
TRIG_IN — R8 1k 1W — TRIG_IN_LED
U4 FOD817A opto (TRIG_IN): A, K, C, E; TRIG_IN_LED — GND; TRIG_IN_GPIO — GND
+3V3 — R9 10k pull-up — TRIG_IN_GPIO
D3 1N4148 (LED reverse protect): GND — A — K — TRIG_IN_LED

TRIGGER OUT: open-collector NPN, pull-up rail chosen by jumper

TRIG_OUT_GPIO — R10 2.2k (base) — TRIG_OUT_BASE
TRIG_OUT_BASE — R11 10k (base-emitter) — GND
Q1 2N3904 NPN (TO-92, pin 1 E, 2 B, 3 C): B — TRIG_OUT_BASE; C — TRIG_OUT; E — GND
TRIG_PU — R12 1.5k pull-up — TRIG_OUT
J9 TRIG_OUT pull-up sel.: 3V3, TRIG_PU, +5V, TRIG_PU, +12V, TRIG_PU → 3V3, PU, 5V, PU, 12V, PU

Jumper a row (3V3 / 5V / 12V) to pull TRIG_OUT up to that rail; remove it if the receiver has its own pull-up.
Pull-up current: 3.3 V ~2 mA, 5 V ~3 mA, 12 V ~7 mA (enough for a 6.5 mA opto input such as MPS4200).
24 V receivers: no jumper, add an external 24 V pull-up in the harness. 3V3 for eSync 2 high-Z ports (3.3 V max).

NOTES
1. ONE common GND (Pico, sensors, servo PSU). SBUS needs a shared ground reference.
   RM2 WiFi breakout (off-board, cabled to the Pico Plus 2 SP/CE connector): uses GP32-GP36 only, no overlap with I2C (GP4/5) or any other signal.
2. Strain gauges: SEPARATE board (kicad/gaugeboard), reached by a 4-wire I2C loom (GND, +5V, SDA, SCL; 0.1in headers, keyed plugs).
   That board makes its own clean 3.3 V from +5V and holds both ADS122C04s, the gauge headers and the filtering.
   ADS #2 needs its ADDR0 jumper bridged (expected address 0x41, unconfirmed; firmware scans at boot). I2C pull-ups come from the ADS122C04 breakouts on the gauge board and from the IMU breakout.
3. SBUS: GP20 (TX) and GP21 (RX) on one net (UART inverted in firmware). One 3x6 servo header (6 plugs of GND, V, SIG; footprint HastaDAQ:PinHeader_3x06): V = VSERVO (8.4 V), SIG = SBUS (all six in parallel), GND.
4. POWER: 12 V in (no fuse; the bench supply's current limit is the protection) -> +12V rail -> (a) Pololu D42V55F8 = VSERVO 8.4 V (servos), (b) SparkFun BabyBuck 5V = +5V (Orbis, gauge-board loom, Schottky -> Pico VSYS). VSYS max is 5.5 V. USB (VBUS) is left for programming; with USB only, +5V is not live.
   BabyBuck EN is left open (its own divider enables it). 5V bulk: module has 2x22uF; 100uF added near the Pico/ports.
5. Triggers: TRIG_IN accepts 3.3-24 V (opto). TRIG_OUT is open collector with a jumper-selected pull-up. Pin order: see each symbol.
   Main supply cable (2x4, pins 1-8 row-major): 1,8 VSUPPLY (12 V) | 2,7 VGND | 3 TRIG_IN | 6 TRIG_OUT | 4 SOFTSTOP+ | 5 SOFTSTOP-.
   SOFTSTOP+ -> 1k -> GP16 (10k pull-up to 3V3, 5.1 V zener to GND). SOFTSTOP- = GND: loop closed = run, broken = stop.
   The 2x4 connectors (this one and ENC1-4) are plain 0.1in headers for now; use keyed/shrouded plugs: a rotated plug would put 5V on a CS pin. Measured system current was <= 1 A at 8.4 V, well inside ~3 A per 0.1in pin.
6. Encoders: 4 x 2x4 headers (pins 1-8: GND, CS, SCK, MOSI, MISO, 3V3, 5V, GND) on one SPI1 bus; CS = GP13/GP9/GP8/GP7.
   MT6835 uses 3V3 (3.3 V logic); RLS Orbis uses 5V with 3.3 V logic. Check MISO tri-states when CS is high.
   The four ENC headers are a generic SPI port (SPI1: SCK GP10, MOSI GP11, MISO GP12, own CS each, 3V3 and 5V supply), so any 3.3 V SPI device can be plugged in, not just encoders. Each device needs its own CS and an MISO that tri-states when CS is high.

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